

LACKLAND AFB, TX, USA

WMO: 722535

Lat: 29.383N

Lon: 98.583W

Elev: 690

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.7	33.0	12.5	10.7	42.7	17.1	13.4	45.7	25.5	56.0	22.2	54.2	6.8	340	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.2	100.5	74.2	99.1	74.2	97.1	74.2	79.9	88.1	78.8	87.2	78.0	86.7	10.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	149.8	82.0	77.0	144.2	81.6	75.7	137.8	80.8	43.9	88.4	42.8	86.7	41.8	86.8	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.9	17.2	DB	21.3	104.3	6.8	2.6	16.4	106.1	12.4	107.6	8.6	109.1	3.6	111.0
			WB	19.8	80.8	6.1	1.5	15.4	81.9	11.8	82.8	8.3	83.6	3.8	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.5	53.0	57.7	63.2	70.5	78.1	83.4	85.3	85.8	80.5	71.6	61.6	54.3	(6)
	DBStd	13.74	9.02	9.81	8.63	7.21	5.75	4.02	3.29	3.11	5.20	7.71	8.91	8.93	(7)
	HDD50	190	72	37	11	1	0	0	0	0	0	1	14	54	(8)
	HDD65	1338	382	240	137	30	2	0	0	0	1	31	166	349	(9)
	CDD50	7668	166	251	421	614	870	1001	1096	1111	916	672	362	188	(10)
	CDD65	3342	11	35	82	194	407	551	631	646	467	237	63	18	(11)
	CDH74	36862	78	247	545	1629	3976	6461	7900	8235	5118	2159	416	98	(12)
	CDH80	17933	6	55	119	553	1726	3297	4334	4593	2454	727	63	6	(13)
(14) Wind	WSAvg	8.1	7.4	8.2	8.9	9.5	9.5	9.0	8.3	7.7	7.0	7.6	7.3	7.1	(14)
(15) Precipitation	PrecAvg	30.4	1.9	1.7	2.3	2.6	3.9	3.3	2.6	2.0	3.1	3.1	1.9	1.9	(15)
	PrecMax	49.8	5.2	6.6	7.1	7.2	9.1	9.8	12.0	8.0	8.9	10.9	8.7	13.2	(16)
	PrecMin	12.4	0.0	0.1	0.0	0.3	0.2	0.2	0.1	0.0	0.2	0.0	0.0	0.2	(17)
	PrecStd	10.5	1.5	1.6	1.5	1.9	2.6	2.5	3.2	2.3	2.2	3.1	2.2	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.4	86.2	88.1	94.8	98.7	102.2	102.8	102.5	100.0	93.2	84.6	80.8	(19)
		MCWB	64.8	63.1	65.6	69.3	73.9	75.0	73.8	74.1	72.8	72.2	70.2	66.7	(20)
	2%	DB	75.7	80.6	83.1	89.8	95.1	99.1	100.3	100.4	96.8	89.9	81.7	76.9	(21)
		MCWB	61.7	63.8	66.5	71.0	74.3	75.1	73.8	74.4	73.4	72.7	67.9	64.7	(22)
	5%	DB	72.5	76.7	79.9	86.0	91.9	96.6	98.6	98.9	94.2	87.0	79.0	73.1	(23)
		MCWB	61.5	63.3	66.5	69.3	74.2	74.6	73.9	74.5	73.2	71.2	66.7	64.4	(24)
	10%	DB	69.0	72.6	76.7	82.5	89.1	93.4	96.2	96.9	91.2	84.1	75.6	70.1	(25)
		MCWB	60.0	62.5	65.0	68.6	73.6	74.9	74.4	74.6	73.4	70.3	66.6	62.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	71.1	73.9	78.8	80.1	81.2	80.5	81.0	80.2	78.3	74.5	72.3	(27)
		MCDB	75.0	76.1	79.3	85.6	85.7	87.9	90.0	93.6	86.6	83.9	78.3	75.7	(28)
	2%	WB	67.9	69.0	71.2	75.7	78.2	79.7	78.5	79.3	78.3	76.7	72.7	69.9	(29)
		MCDB	71.7	73.5	77.2	82.9	86.7	88.0	87.9	88.2	85.5	82.6	77.3	73.7	(30)
	5%	WB	65.4	67.6	69.5	73.4	77.0	78.5	77.8	78.1	77.3	75.4	71.0	67.7	(31)
		MCDB	69.5	72.5	74.8	80.6	86.3	87.5	87.2	86.9	85.0	81.5	75.6	71.2	(32)
	10%	WB	62.5	65.8	68.1	72.1	75.8	77.5	77.1	77.3	76.4	74.1	69.1	64.5	(33)
		MCDB	66.6	70.6	73.6	78.8	84.5	87.0	86.6	86.1	84.4	80.3	73.9	68.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.6	21.9	21.2	21.4	19.5	19.5	20.7	21.2	20.2	21.0	21.1	21.8	(35)
		MCDBR	27.7	26.2	23.6	24.6	22.3	23.0	24.2	24.0	23.1	21.5	22.3	25.3	(36)
	5% WB	MCWBR	16.9	14.8	11.9	10.8	6.8	5.2	5.1	5.2	5.9	8.3	12.0	15.5	(37)
		MCWBR	20.1	19.1	18.2	18.3	18.7	18.8	19.9	19.9	17.8	16.0	17.3	18.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.327	0.334	0.358	0.400	0.446	0.429	0.435	0.421	0.414	0.374	0.360	0.319	(40)
		taud	2.499	2.482	2.431	2.327	2.245	2.348	2.362	2.385	2.355	2.457	2.430	2.532	(41)
	Ebn at Noon		289	296	295	284	270	273	271	274	273	279	274	287	(42)
		Edh at Noon	29	32	36	41	44	40	39	38	38	32	30	27	(43)
(44) All-Sky Solar Radiation	RadAvg		961	1130	1426	1693	1828	2095	2110	2020	1655	1372	1032	854	(44)
	RadStd		108	202	148	171	191	137	184	115	166	182	116	111	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158

Nomenclature: See separate page